

# Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Beat Optical, and Does It Hold Across ViT and CNN?

Anonymous WACV Applications Track submission

Paper ID \*\*\*\*\*

## Abstract

001 Detecting “dark” vessels—ships not broadcasting their  
002 Automatic Identification System (AIS) signal—in  
003 spaceborne synthetic aperture radar (SAR) is opera-  
004 tionally important and label-scarce. Pretrained back-  
005 bones can help, but it is unclear which pretrain-  
006 ing recipe transfers best, and whether the answer  
007 is the same for Transformer and convolutional back-  
008 bones. We propose a controlled label-efficiency study  
009 on xView3-SAR [8] that holds one point-native detec-  
010 tor fixed and varies only the downloaded backbone ini-  
011 tialization, across two size-matched tracks (ViT-B/16,  
012 ~86M; ConvNeXt-V2-Base, ~89M) and four pretrain-  
013 ing recipes each: random, optical, SAR, and super-  
014 vised out-of-domain SAR. Because every backbone is a  
015 released checkpoint, differences are attributable to the  
016 pretraining source rather than to our own optimization  
017 choices.

## 018 1. Introduction

019 Detecting ships not broadcasting their AIS signature—  
020 “dark vessels”—in spaceborne SAR is a critical  
021 maritime-monitoring problem with scarce labels:  
022 xView3-SAR [8] provides Sentinel-1 imagery, but  
023 its human-verified dark-vessel labels are few, so  
024 supervised-only training overfits.

025 Pretrained backbones can help, but it is un-  
026 clear which pretraining recipe transfers best: self-  
027 supervision on generic optical satellite imagery (fMoW-  
028 RGB [2] via SatDINO [9]), self-supervision on large-  
029 scale SAR (SAR-1M via SARMAE [7]), or super-  
030 vised detection on labeled out-of-domain SAR (LS-  
031 SSDD [11])—and whether the best recipe is the same  
032 for a ViT [4] and a ConvNeXt [10]. We compare re-  
033 leased checkpoints rather than pretraining our own so  
034 that each arm faithfully represents its recipe as pub-

lished; re-pretraining would confound the pretraining  
source with our own optimization choices [13].

With the detector, schedule, and input fixed, we ask:  
(Q1) how much labeled xView3 data does each recipe  
save relative to random initialization to reach a tar-  
get dark-vessel F1; (Q2) does SAR-domain pretraining  
beat optical in the scarce-label regime; and (Q3) is that  
ordering architecture-general, holding for both a ViT  
and a size-matched CNN? We hypothesize that, within  
each track, SAR-domain pretraining yields the steep-  
est low-label gains (random < optical  $\lesssim$  supervised <  
SAR), and that this ordering holds for both architec-  
tures.

## 2. Approach

Design. The study has two architecture-matched  
tracks—ViT-B/16 (~86M) and ConvNeXt-V2-Base  
(~89M, deliberately size-matched)—each with four  
arms that share the same head, optimizer, schedule,  
and seeds; only the downloaded initialization differs  
(Table 1). Initialization is the only variable within a  
track and architecture the only variable across matched  
roles, so comparing arm  $k$  to arm  $k+4$  isolates the ar-  
chitecture family with the pretraining role held fixed.

Pretraining regimes. The recipes differ in modality,  
labels, and domain: optical (SatDINO, DINO [1];  
BigEarthNet-S2) is a different sensing modality (no  
speckle, different scattering physics); SAR, self-  
supervised (SARMAE, an MAE [5]; BigEarthNet-S1)  
is the same modality but pretrained without detection  
labels on a different scene distribution; and labeled out-  
of-domain SAR (LS-SSDD) is the same modality and  
a related supervised task, but a different acquisition.  
Q2 thus contrasts “same pixels, different task” against  
“different pixels, same task.”

| Arm | Track | Backbone          | Initialization                | Params   | On curve? | Role                       |
|-----|-------|-------------------|-------------------------------|----------|-----------|----------------------------|
| 1   | ViT   | ViT-B/16          | random init                   | ~86M     | yes       | ViT floor                  |
| 2   | ViT   | ViT-B/16          | SatDINO (fMoW-RGB, DINO)      | ~86M     | yes       | optical-domain transfer    |
| 3   | ViT   | ViT-B/16          | SARMAE (SAR-1M, MAE)          | ~86M     | yes       | SAR-domain transfer        |
| 4   | ViT   | ViT-B/16          | LS-SSDD supervised (we train) | ~86M     | yes       | labeled out-of-domain SAR  |
| 5   | CNN   | ConvNeXt-V2-B     | random init                   | ~89M     | yes       | CNN floor                  |
| 6   | CNN   | ConvNeXt-V2-B     | BigEarthNet-S2 (optical RS)   | ~89M     | yes       | optical-domain transfer    |
| 7   | CNN   | ConvNeXt-V2-B     | BigEarthNet-S1 (SAR RS)       | ~89M     | yes       | SAR-domain transfer        |
| 8   | CNN   | ConvNeXt-V2-B     | LS-SSDD supervised (we train) | ~89M     | yes       | labeled out-of-domain SAR  |
| R1  | ref   | ConvNeXt-V2-B     | ImageNet (contingent)         | ~89M     | no        | generic-transfer reference |
| R2  | ref   | YOLO26            | COCO-pretrained               | external | no        | detector reference         |
| R3  | ref   | LocateAnything-3B | zero-shot                     | ~3B      | no        | VLM reference              |

Table 1. Eight controlled arms in two size-matched tracks (~86M vs. ~89M), plus three external references reported off the curves. Optical anchors are SatDINO [9] (ViT) and BigEarthNet-S2 [3] (CNN); SAR anchors are SARMAE [7] (ViT) and BigEarthNet-S1 [3] (CNN).

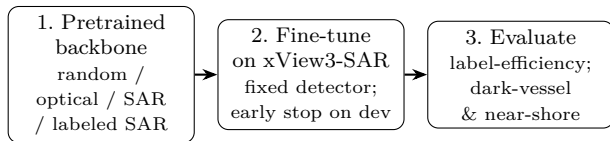


Figure 1. The three stages, run identically for every arm; only stage 1 (the downloaded backbone) varies within a track.

Detector. xView3 labels are points and its metric matches predictions by distance, not box IoU, so we use a point-native heatmap head in the style of CenterNet / Objects as Points [12]: each vessel is a Gaussian blob on a stride-4 map, decoded by peak-finding with distance-based non-maximum suppression. Every arm receives the same fixed 3-channel input [VH, VV, VH-VV] in dB, so input handling is never a confound between arms.

Fine-tuning and comparability. All arms fine-tune end-to-end through one code path with shared AdamW, augmentation, and seeds, and are trained until dev-split F1 stops improving (early stopping), not for a fixed epoch budget; the decision threshold is selected on dev and frozen before test. Continuous-integration guard tests make comparability a checked invariant rather than a claim: a backbone-parity test asserts identical within-track parameter counts and matching head-adaptor geometry, a checkpoint-load test asserts each backbone loads its pretrained weights (not silently random), and a scorer-immutability test freezes the evaluation code (Figure 1).

### 3. Data and Evaluation

We use xView3-SAR [8] via the SARFish mirror (Sentinel-1 GRD, dual-polarization VH/VV): 75–150 training scenes chipped at  $800 \times 800$  ( $\sim 10^4$  chips), sub-sampled at the scene level to 10/25/50/100% of labels and split 75/15/10 (train/dev/test); the ~50 human-verified scenes are a disjoint, once-touched final-evaluation set. Arms 4 and 8 pretrain on LS-SSDD-v1.0 [11] (~9,000 labeled  $800 \times 800$  sub-images), used only as a pretraining source and never mixed into the xView3 splits. The headline result is a two-track label-efficiency curve (detection F1 vs. 10/25/50/100% labels; eight lines) plus dark-vessel recall and near-shore F1, the operationally hardest slices: the low-label gap over the random floor answers Q1–Q2, and matched-role gaps across tracks answer Q3. External references (YOLO26 [6], a zero-shot VLM, and a contingent ImageNet-ConvNeXt) situate the eight arms without entering the controlled comparison.

### 4. Timeline and Feasibility

Two contributors work independent lanes (data/scorer; detector/eval) with no pretraining handoff to block on. Because all foundation-model backbones are downloaded, compute is modest: the full eight-arm grid with seed reruns plus the two small LS-SSDD supervised backbones is ~293–360 GPU-hours (~3–4 nights on one  $8 \times V100$  node)—we download checkpoints for comparability of recipes as published, and the low cost is a side benefit. The downloaded-backbone arms run first to de-risk the fixed-input channel format. We target the WACV 2027 Round-2 deadline (paper registration August 21, 2026; submission August 28, 2026, AoE).

## References

- [1] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In IEEE/CVF International Conference on Computer Vision (ICCV), 2021. 1
- [2] Gordon Christie, Neil Fendley, James Wilson, and Ryan Mukherjee. Functional map of the world. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2018. 1
- [3] Kai Norman Clasen, Leonard Hackel, Tom Burgert, Gencer Sumbul, Begüm Demir, and Volker Markl. reBEN: Refined BigEarthNet dataset for remote sensing image analysis. In IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025. arXiv:2407.03653. 2
- [4] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In International Conference on Learning Representations (ICLR), 2021. 1
- [5] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. 1
- [6] Glenn Jocher et al. Ultralytics YOLO26. <https://github.com/ultralytics/ultralytics>, 2026. 2
- [7] Danxu Liu, Di Wang, Hebaixu Wang, Haoyang Chen, Wentao Jiang, Yilin Cheng, Haonan Guo, Wei Cui, and Jing Zhang. SARMAE: Masked autoencoder for SAR representation learning. arXiv preprint arXiv:2512.16635 (to appear, CVPR), 2026. 1, 2
- [8] Fernando Paolo, Tsu-ting Tim Lin, Ritwik Gupta, Bryce Goodman, Nirav Patel, Daniel Kuster, David Kroodsma, and Jared Dunnmon. xView3-SAR: Detecting dark fishing activity using synthetic aperture radar imagery. In Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2022. 1, 2
- [9] Jakub Straka and Ivan Gruber. SatDINO: A deep dive into self-supervised pretraining for remote sensing. arXiv preprint arXiv:2508.21402, 2025. 1, 2
- [10] Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. 1
- [11] Tianwen Zhang, Xiaoling Zhang, Xiao Ke, Xu Zhan, Jun Shi, Shunjun Wei, Dece Pan, Jianwei Li, Hao Su, Yue Zhou, and Feng Xu. LS-SSDD-v1.0: A deep learning dataset dedicated to small ship detection from large-scale Sentinel-1 SAR images. Remote Sensing, 12 (18):2997, 2020. 1, 2
- [12] Xingyi Zhou, Dequan Wang, and Philipp Krähenbühl. Objects as points. arXiv preprint arXiv:1904.07850, 2019. 2
- [13] Barret Zoph, Golnaz Ghiasi, Tsung-Yi Lin, Yin Cui, Hanxiao Liu, Ekin D. Cubuk, and Quoc V. Le. Re-thinking pre-training and self-training. In Advances in Neural Information Processing Systems (NeurIPS), 2020. 1